

Project: Prism IQ

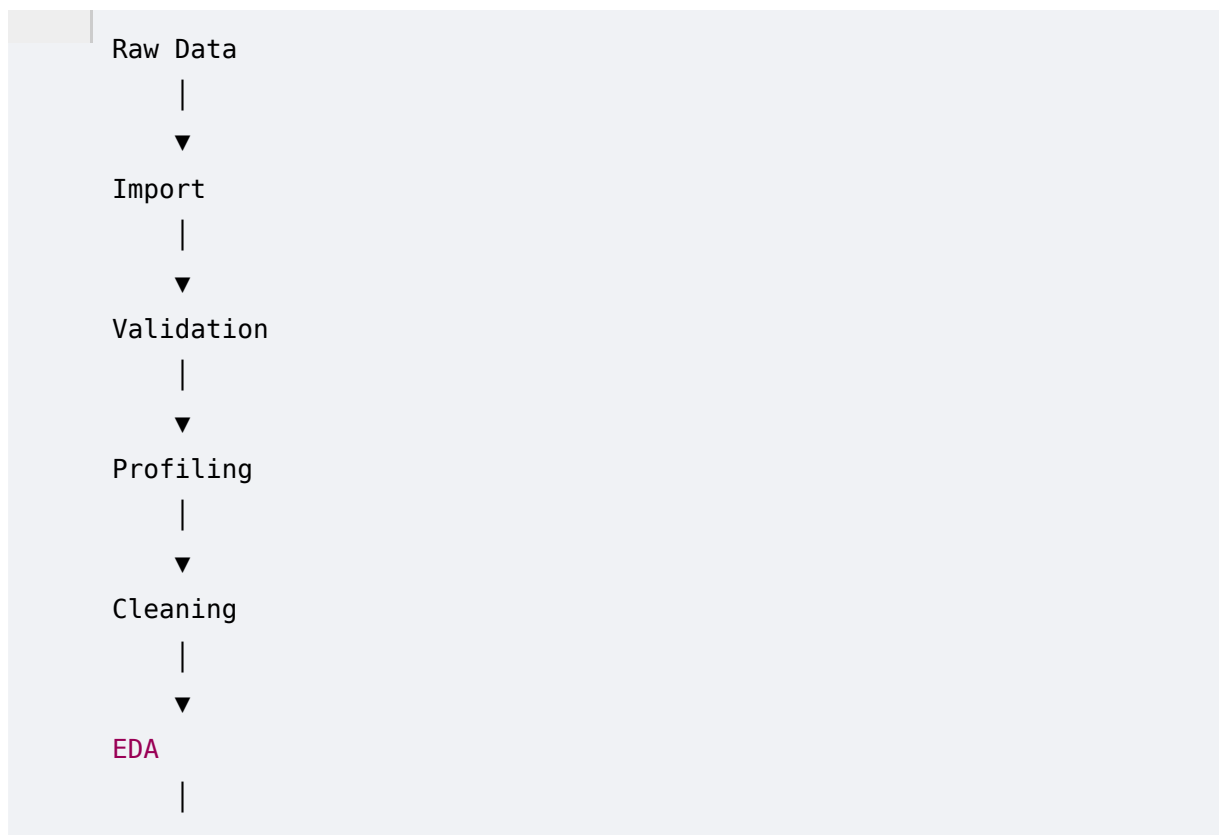
Version: 1.0

Status: Architecture Freeze

1. System Overview

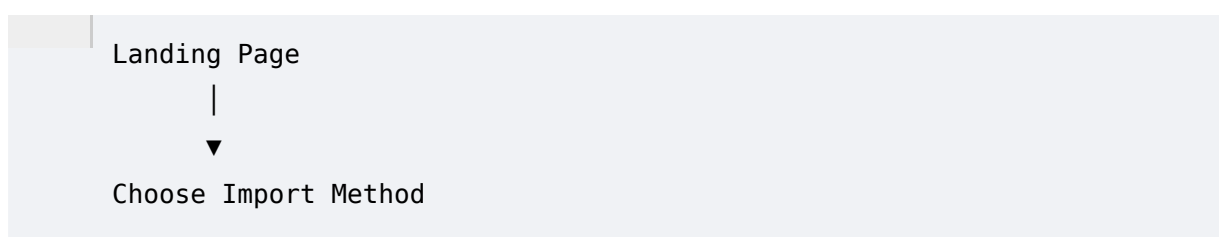
Prism IQ follows a **pipeline-driven architecture**, where every dataset passes through a fixed sequence of processing stages.

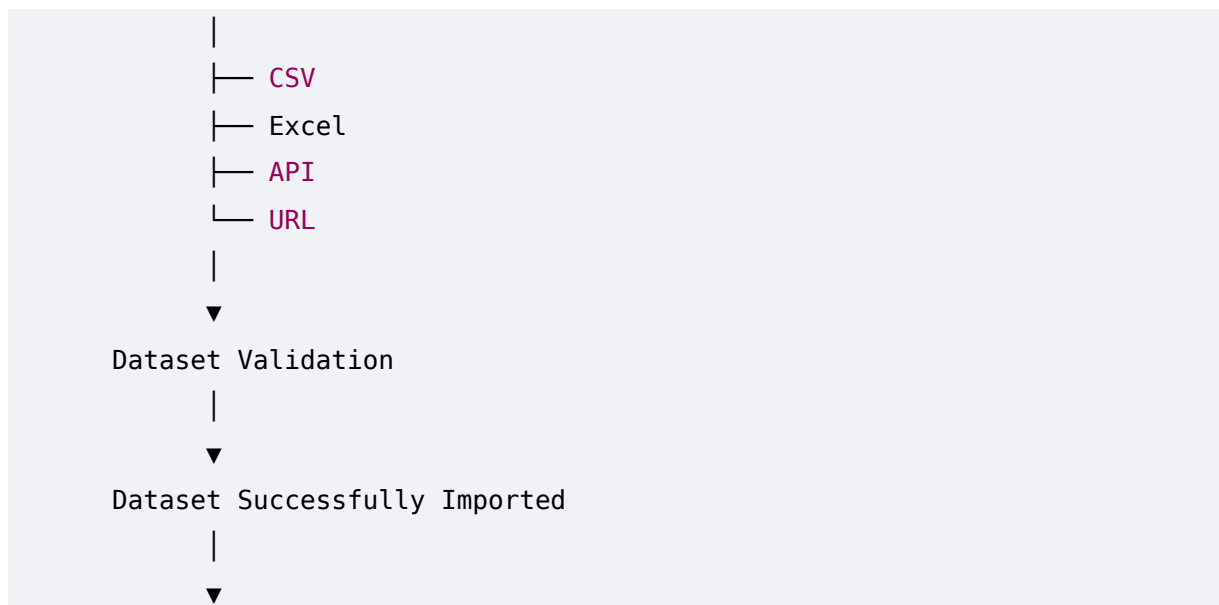
Instead of treating Upload, Cleaning, EDA, Machine Learning, and Reporting as separate features, they are connected into a single intelligent workflow.



This ensures that every stage builds upon the validated output of the previous stage.

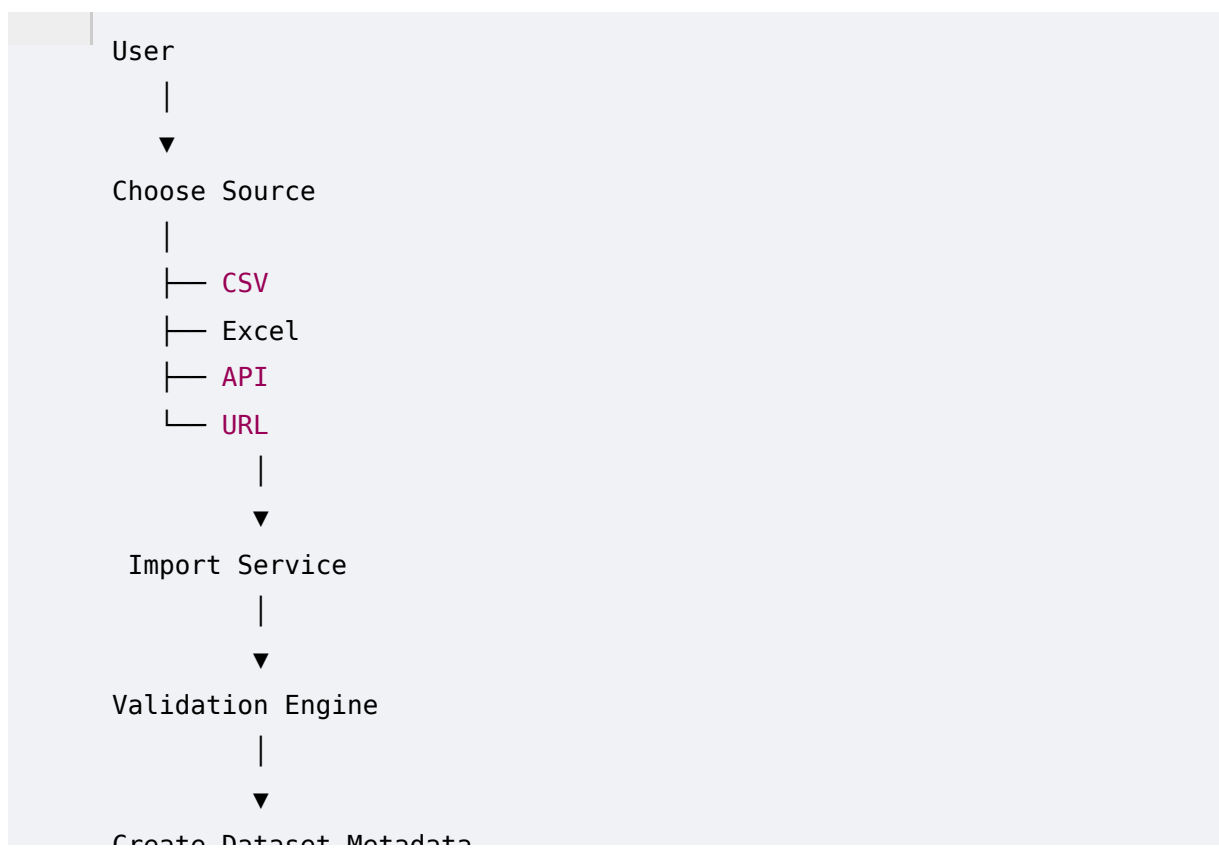
2. Complete User Journey





3. Data Import Flow

Prism IQ supports four import methods.



Validation Rules

Every imported dataset is validated before entering the analytics pipeline.

Validation checks include:

- Supported file format

- File size
- Empty dataset
- Duplicate column names
- Invalid headers
- Unsupported data types
- API response validation
- HTML table extraction success

If validation fails, the user receives descriptive error messages without stopping the application.

4. Data Profiling Flow

Immediately after successful import, profiling begins automatically.

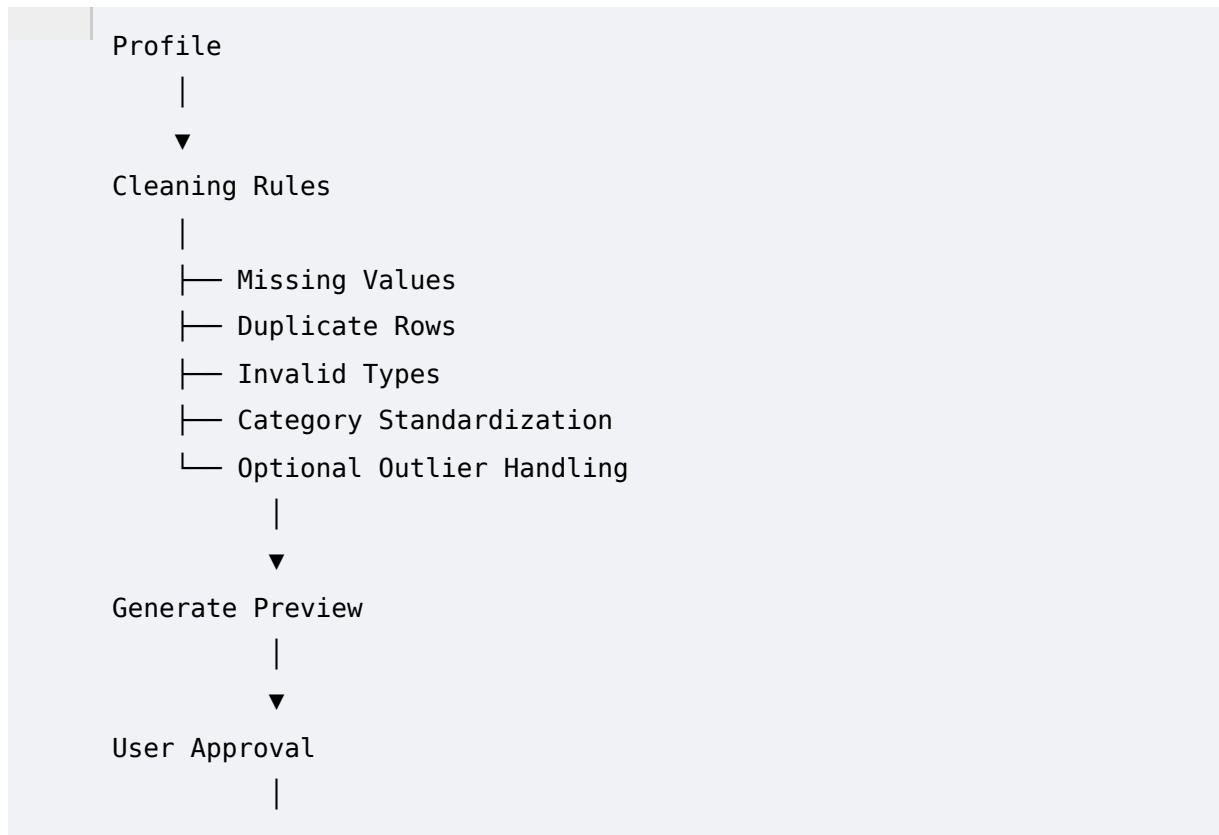


Output:

- Dataset Summary
 - Quality Score
 - Column Summary
 - Initial Insights
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5. Smart Cleaning Flow

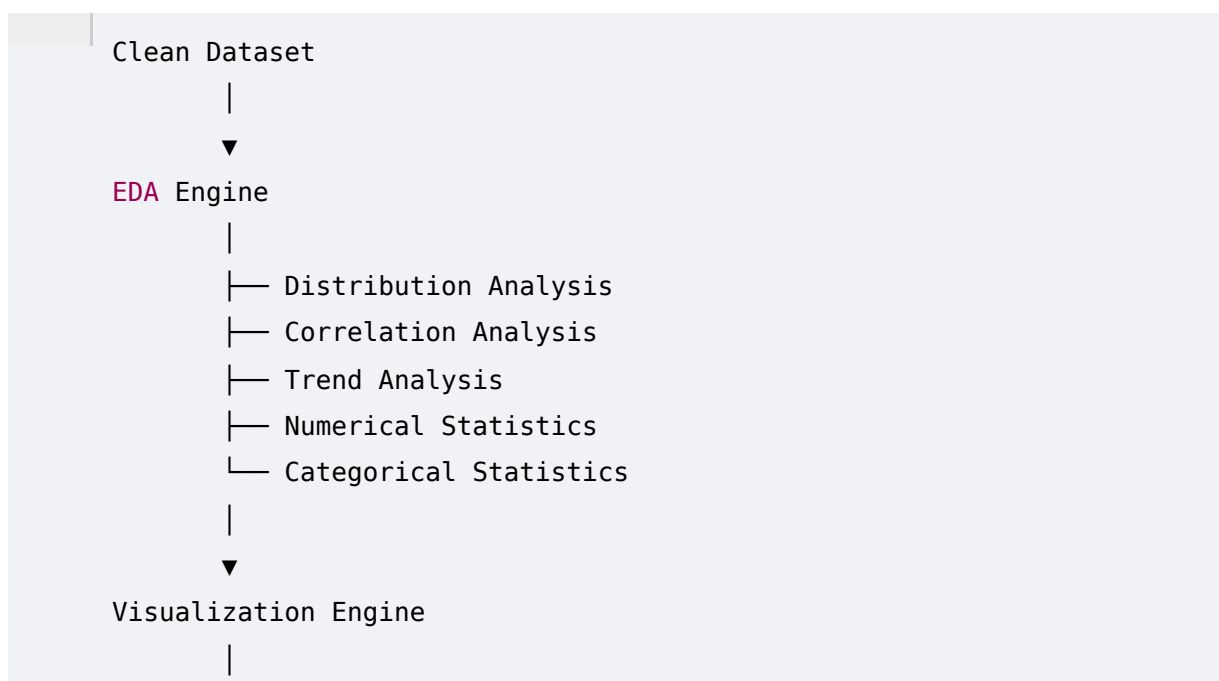
The cleaning engine performs rule-based preprocessing.



The original dataset always remains unchanged.

The cleaned version is stored separately.

6. Exploratory Data Analysis Flow



▼ Interactive Dashboard

Generated visualizations include:

- Histogram
- Scatter Plot
- Bar Chart
- Line Chart
- Box Plot
- Heatmap
- Correlation Matrix

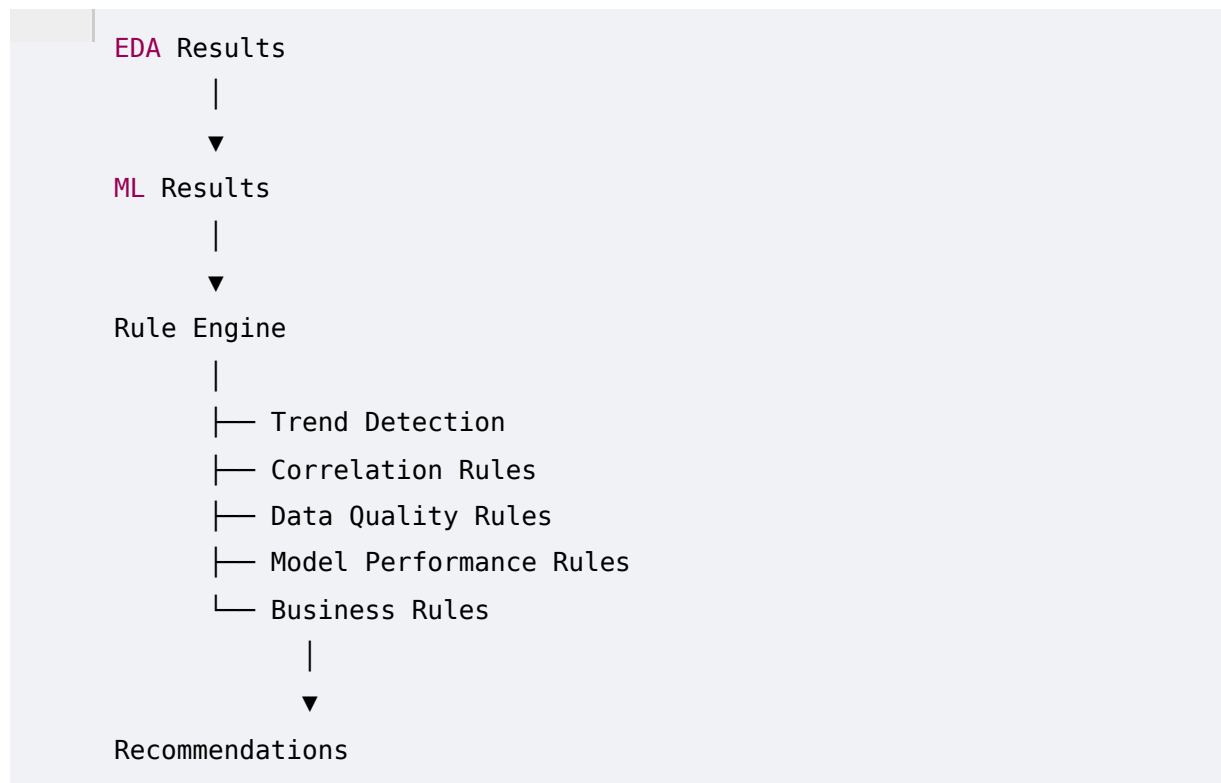
7. Predictive Analytics Flow



The platform automatically compares supported models and presents the best-performing one along with evaluation metrics.

8. Recommendation Engine Flow

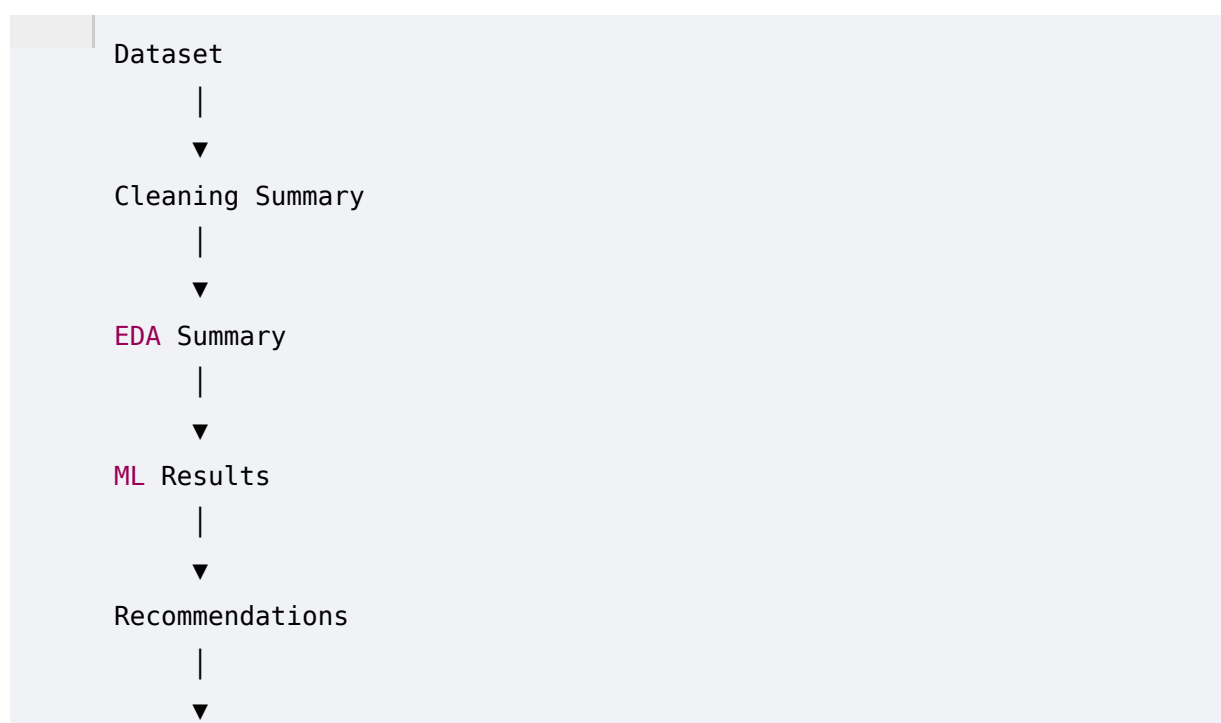
Unlike LLM-based systems, the MVP uses a deterministic rule engine.

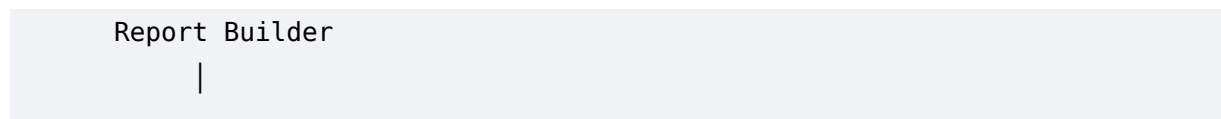


Example outputs:

- High-performing product categories
- Customer segments requiring attention
- Strong positive or negative correlations
- Suggested business actions

9. Report Generation Flow





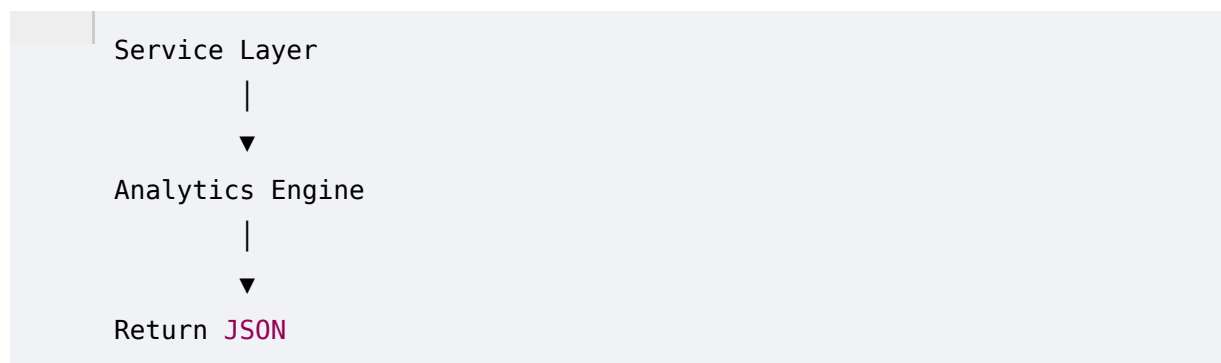
Each report includes:

- Executive Summary
- Dataset Overview
- Cleaning Summary
- Key Visualizations
- Model Performance
- Recommendations
- Conclusion

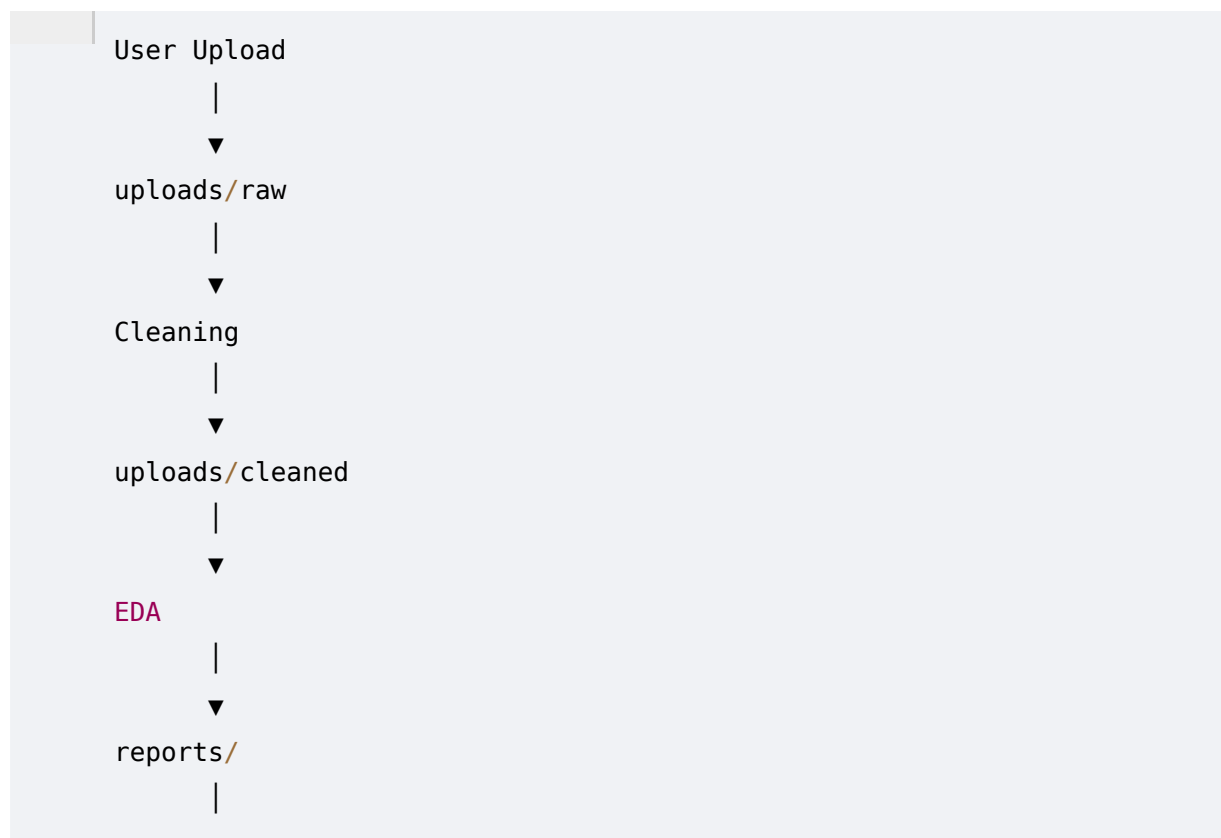
10. Frontend–Backend Communication Flow



For analytical operations:

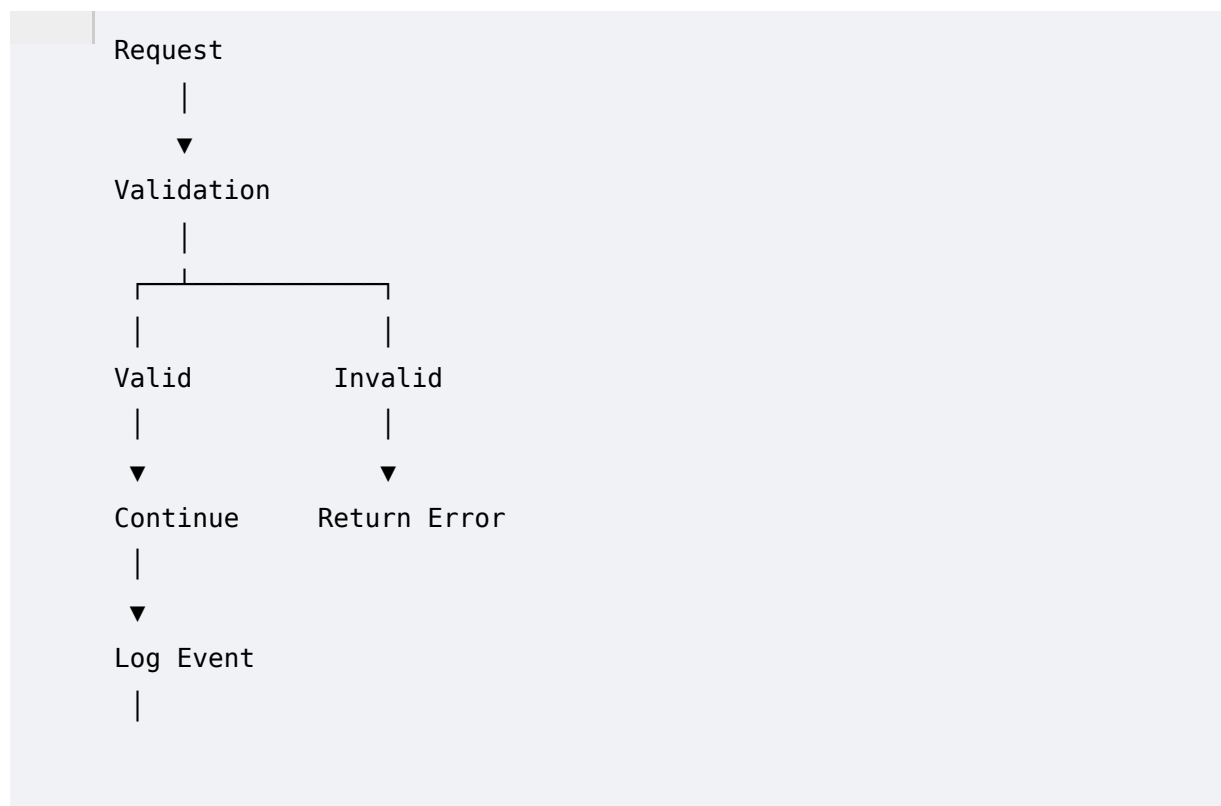


11. File Management Flow



Only metadata is stored in PostgreSQL.

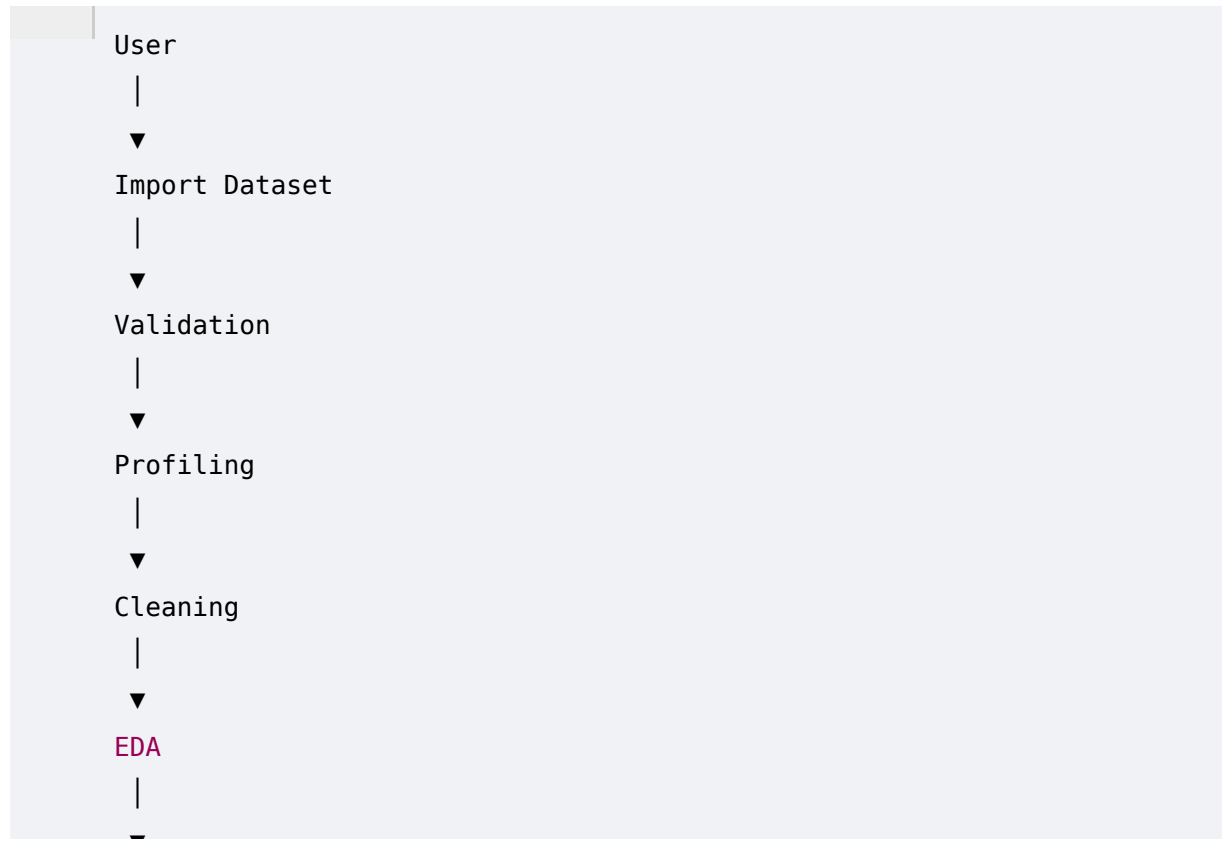
12. Error Handling Flow



▼

Every API returns a standardized response format with status, message, data, and errors where applicable.

13. End-to-End System Workflow

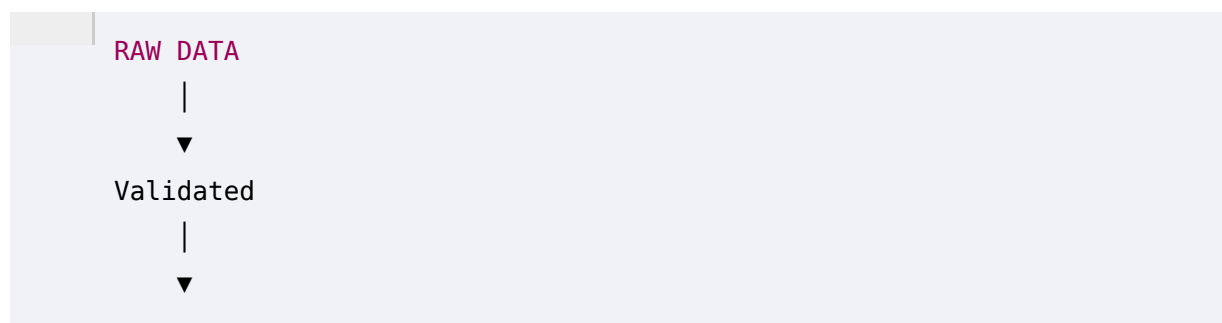


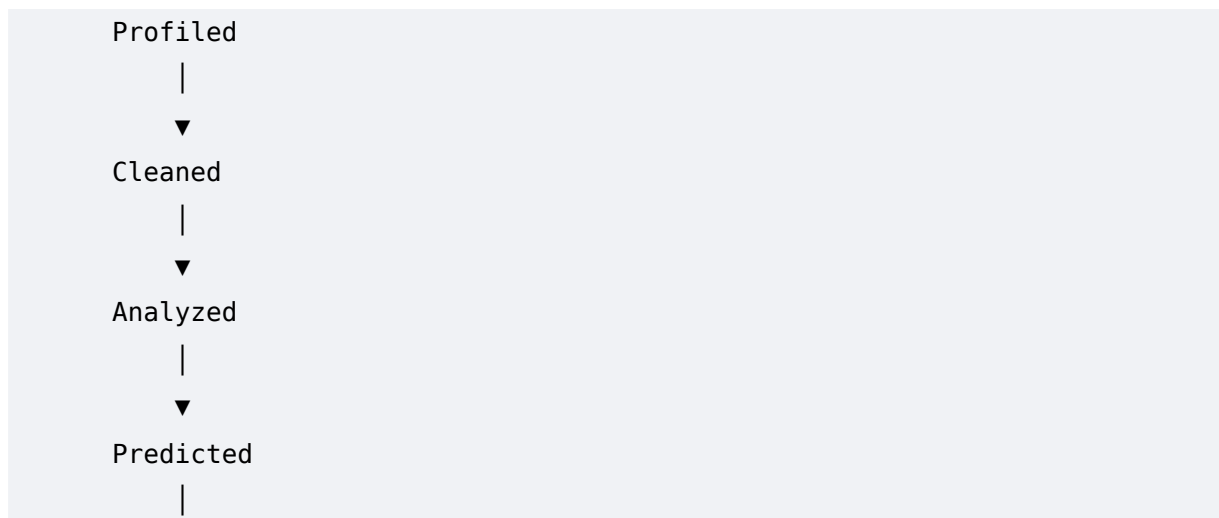
14. Background Processing Strategy (Future-Ready)

Current MVP executes processing synchronously for simplicity.

The architecture is designed so that long-running tasks (large dataset profiling, model training, report generation) can later be moved to asynchronous workers such as Celery + Redis without major architectural changes.

15. Data Lifecycle





Every stage updates the dataset status, making progress traceable and simplifying debugging.

16. Design Principles

- **Single Source of Truth:** Dataset metadata lives in PostgreSQL.
 - **Immutable Raw Data:** Original uploads are never modified.
 - **Pipeline Consistency:** Each stage consumes the output of the previous stage.
 - **Loose Coupling:** Import, Cleaning, EDA, ML, and Reporting are independent modules.
 - **Extensibility:** New importers, models, or report templates can be added without changing existing flows.
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Phase 5 Deliverables

By the end of this phase, the complete operational flow of Prism IQ is frozen:

- End-to-end user journey
- Data processing workflow
- Frontend ↔ Backend communication
- Machine learning pipeline
- Report generation flow
- Error handling strategy
- File lifecycle
- Future-ready processing architecture